Related Work: SkillTree Delivery, Security Posture, Reporting, and Open Science

SkillTree delivery as prerequisite learning and gamified practice context I

SkillTree supplies the delivery substrate: Project, Subject, Skill, learning-path dependencies, quizzes, and report endpoints (National Security Agency 2020, 2026b; SkillTree 2026). Active Skillference uses those concepts directly rather than inventing a new learning platform. The graph layer adds prerequisite validation and connectivity checks because the learning object is the ordered dependency structure, not only the text attached to each node (Bloom 1956; Bloom 1968; Liang et al. 2018; Doignon and Falmagne 1985; Falmagne and Doignon 2011).

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Gamification is relevant but not itself the contribution; SkillTree provides the gamified runtime, and the project contributes the verified content and graph (Dicheva et al. 2015). Visual skill-tree and gamified microlearning sources provide useful delivery precedent for making dependencies and progress inspectable, but they do not constitute learner-outcome evidence for this export (Burau 2022; Septiani and Rosmansyah 2021). Constructive alignment and formative-assessment sources supply the education-facing language for this design: objectives, activities, feedback, and quiz gates are treated as aligned checkpoints, not as outcome evidence (Biggs 1996; Bloom 1968; Black and Wiliam 1998; Shute 2008).

SkillTree provenance, deployment security, and access-control boundaries I

SkillTree's origin matters for the delivery argument. NSA describes SkillTree as an internally developed open-source training platform, and the public `skills-service` repository and NSA open-source portal identify the service as public government-origin software rather than a generic third-party widget (National Security Agency 2020, 2026b, 2026a). Active Skillference therefore treats SkillTree as a real delivery substrate with explicit platform ownership boundaries: SkillTree owns learner progress, quiz state, authentication, ranking and reporting semantics, while this project owns the curriculum graph, provenance gates, manuscript evidence and exported SkillTree payload.

That provenance raises the bar for deployment language. A national-security posture would still require organization-specific threat modeling, zero-trust access decisions, supply-chain provenance, software-development process, artifact signing, SBOM visibility, and adversary-technique coverage beyond this educational repository (Rose et al. 2020; National Institute of Standards and Technology 2022; SLSA Framework 2026; Sigstore 2026; Cybersecurity and Infrastructure Security Agency 2026; MITRE 2026). Active Skillference can support defensible training review because claims, figures, source roles, dashboard artifacts and SkillTree handoff routes are inspectable; it does not certify operator competence, harden a production deployment, replace APT-aware threat modeling, or imply NSA approval of the curriculum.

Reproducible reporting, accessible artifacts, and source-role auditing I

The content-provenance binding is closest in spirit to literate programming and literate statistical reporting: code and explanation are kept mutually inspectable, and R/knitr- style workflows bind prose to executable analysis so reported numbers can be regenerated (Knuth 1984; Xie 2015). Peng's reproducible-research argument is the reporting boundary used here: when a claim is computational, readers should be able to rerun the workflow that produced it (Peng 2011; Sandve et al. 2013). Scientific-computing practice adds the software-maintenance side of the same argument: versioned code, automated checks, and tested workflows are part of the scholarly surface when the artifact is executable (Wilson et al. 2014). W3C PROV gives the formal provenance vocabulary for entities, activities, agents, derivations, bundles, and

validity constraints, while reproducible-science sources keep the claim boundary tied to transparent methods rather than trust language (Groth and Moreau 2013; Moreau and Missier 2013; Cheney et al. 2013; Munafò et al. 2017). FAIR and transparency guidelines extend that boundary from rerunnable computation to metadata and open-science claim discipline (Wilkinson et al. 2016; Nosek et al. 2015). Active Skillference differs by keeping a central claim registry, applying the same gate to curriculum content and manuscript prose, and refusing export when any token is unbacked. The DAG engine uses standard graph algorithms for cycle detection and topological ordering (Cormen et al. 2022). Figures use an accessible categorical palette rather than a single-hue theme (Wong 2011). The figure-registry validator also treats alt text and semantic caption metadata as reporting obligations, aligning the

artifact layer with ACM/SIGCHI, Section 508 and WCAG guidance that accessible figures and web content should carry meaningful alternatives rather than empty labels (ACM DIS 2023 Accessibility Chairs 2023; Section508.gov 2026; World Wide Web Consortium 2023).

Release metadata and license notices are treated as delivery evidence, not as scientific or learner-efficacy support. Software citation principles, Citation File Format and CodeMeta guide the citation and metadata surfaces (Smith et al. 2016; Citation File Format Project 2026; CodeMeta Project 2026). GitHub and Zenodo documentation bound the repository and DOI workflow to an actual public target and deposit (GitHub Docs 2026; Zenodo 2026). SPDX, OSI and REUSE license-notice references support the MIT and third-party notice signposting (SPDX Workgroup 2026; Open Source Initiative 2026; Free Software Foundation Europe 2024). These references do not provide legal advice, do not

relicense third-party course materials or vendored browser bundles, and do not turn a static review checkout into a production deployment.

The scholarship audit extends that reproducible-reporting idea from numbers to source coverage. Ordinary citation validators can show that `[@key]` resolves, but not that the key has a stated role in the argument. The source matrix makes that role explicit and keeps the bibliography aligned with the manuscript's actual lanes. The frontier gap report adds a narrower check for newly emphasized source areas: a pedagogy, assessment, formal provenance, workflow provenance, knowledge-space, cognitive-diagnosis, message-passing, predictive-coding, design-based evaluation, open-science, security, accessibility, reproducibility, or course-material source must be cited and tied to an explicit claim-support class and, where needed, a boundary row before it can count as a scholarship upgrade.

Open-science education ecosystem, interoperability, and community handoff I

Active Skillference also sits inside a broader open-science educational ecosystem. The Active Inference Institute publishes a public resource hub with community pathways, verified public repositories, and a documented projection boundary; it also publishes public course infrastructure, a START project for tailored Active Inference research and training curricula, and an institute website that frames education as part of an open-science mission (Active Inference Institute 2026b, 2026e, 2026a, 2026c).

Open-science education ecosystem, interoperability, and community handoff I

The cognitive knowledge-base repository is treated as course-material provenance for generated Learn pointers, not as copied prose or as an endorsement claim (Active Inference Institute 2026d). This project does not replace those efforts; it contributes a provenance-bound SkillTree export, quiz gate, and verifier suite that can interoperate with public learning paths. Recent educational Active Inference scholarship also motivates prepared learning environments and active exploration as topics for future classroom evaluation, while this release limits itself to the inspectable curriculum and delivery contract (Di Paolo et al. 2024).